Adapting Language Models for Medical QA: From Layperson Generation to Clinical Fine-Tuning

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PowerPoint Slides: PowerPoint Slides
Video Presentation: Video Presentation
Code Repository: github.com/chelseanbr/aih-project
Data Source: huggingface.co/datasets/lavita/MedQuAD

April 22, 2025

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Abstract

This project explores the use of large language models (LLMs) for medical question answering (QA) through a two-stage pipeline. We first attempt to generate layperson-friendly multiple-choice questions (MCQs) using a prompting approach with domain-specific QA pairs. Due to challenges in formatting consistency and output quality, we pivot to fine-tuning TinyLLaMA on a clinical MCQ dataset to evaluate lightweight model adaptation. Our work highlights both the potential and limitations of using language models for accessible medical education and QA applications. LLMs show strong clinical reasoning [**?**]. Question entailment is also important [**?**]. Another citation [**?**].

1 Introduction

This is where you introduce your work.

2 Related Work

Discuss previous work here.

3 Methodology

Explain what you did.

4 Results

Show your results here.

5 Conclusion

Wrap it up and suggest future directions.

References

[] Asma Ben Abacha and Dina Demner-Fushman. 2019. A Question-Entailment Approach to Question Answering. *BMC Bioinformatics* 20 (2019). https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3119-4 Article number 511:1–511:23.

[] Karan Singhal, Shekoofeh Azizi, Tien Tu, and et al. 2022. Large Language Models Encode Clinical Knowledge. (2022). https://arxiv.org/abs/2212.13138 arXiv preprint arXiv:2212.13138.

[] Jiean Zhu. 2024. Cura-LLaMA: Evaluating Open-Source Large Language Model's Question Answering Capability on Medical Domain. *Applied and Computational Engineering* 90 (2024), 52–60. doi:10.54254/2755-2721/90/20241725

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